

MAT 534 HW09

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Due 11/12

Problem 1. Describe all finitely generated subgroups of $(\mathbb{Q}, +)$. What are the possible values of the rank of such a subgroup?

Solution. The rank of a finitely generated subgroup of $(\mathbb{Q}, +)$ must be $\boxed{0 \text{ or } 1}$; any finitely generated subgroup of $(\mathbb{Q}, +)$ is either the trivial subgroup (rank 0) or of the form $\langle n/M : n \in \mathbb{Z} \rangle$ for some fixed positive integer M (rank 1). The only thing we need to prove is that a nontrivial finitely generated subgroup must be of the latter form.

Suppose that $H \leq \mathbb{Q}$ is generated by nonzero $r_1, \dots, r_k \in \mathbb{Q}$ and let $M = \text{lcm}(r_1, \dots, r_k)$. Then $r_i = a_i/M$ for some integers a_i such that the ideal in $\mathbb{Z}$ generated by the a_i is the unit ideal (i.e. the a_i have gcd 1). On one hand, expressing r_i as a_i/M implies that the subgroup generated by the r_i is contained in the subgroup generated by $1/M$. On the other hand, since $(a_1, \dots, a_k) = (1)$ as ideals in $\mathbb{Z}$, there is a $\mathbb{Z}$ -linear combination of the a_i/M that equals $1/M$, so $1/M \in \langle r_1, \dots, r_k \rangle$, giving the reverse inclusion. Thus, $H = \langle 1/M \rangle$, meaning every nontrivial finitely generated subgroup is of this form. $\square$

Problem 2. Let R be a principal ideal domain. Show that every submodule of R^d is isomorphic to R^e for some $e \leq d$.

Problem 3. Let R be an integral domain with fraction field F .

- (a) Show that an isomorphism of R -modules $R^d \cong R^e$ induces an isomorphism of F -vector spaces $F^d \cong F^e$.
- (b) Conclude that $R^d \cong R^e$ implies $d = e$.

Problem 4. Let R be a principal ideal domain, and M a nontrivial finitely generated torsion-free R -module.

- (a) Show that if $m_1, m_2 \in M$ are such that $\langle m_1 \rangle \cap \langle m_2 \rangle \neq \{0\}$, then $\langle m_1 \rangle + \langle m_2 \rangle = \langle m \rangle$ for some $m \in M$.
- (b) Show that $M/\langle m \rangle$ is torsion-free if and only if $\langle m \rangle$ is maximal among submodules of this type.
- (c) Show that there exists a nonzero element $m \in M$ such that the R -module $M/\langle m \rangle$ is torsion-free.

Problem 5. Let $T: V \rightarrow V$ be a linear transformation of a finite-dimensional F -vector space. After choosing a basis $v_1, \dots, v_n \in V$, we can represent T by an $n \times n$ matrix A . Show that if A' is the $n \times n$ matrix representing T with respect to another basis $v'_1, \dots, v'_n \in V$, then A' is conjugate to A , in the sense that there exists $B \in GL_n(F)$ with $A' = B^{-1}AB$.

Solution. $\square$

Problem 6. Let V be an F -vector space. Denote by $V^* = \text{Hom}(V, F)$ the **dual vector space** of V , consisting of the linear functionals $\varphi: V \rightarrow F$.

- (a) Show that there is a natural linear transformation $V \rightarrow V^{**}$.

(b) If V is finite-dimensional, show that $V \cong V^{**}$.

Solution. (a) Let $f: V \rightarrow V^{**}$ be the map sending $v \in V$ to the functional $f_v \in V^{**}$ such that $f_v(\varphi) = \varphi(v)$ for $\varphi \in V^*$.

(b)

□

Problem 7. Let V be an F -vector space. For $\varphi \in V^*$ and $v \in V$, let $\langle \varphi, v \rangle = \varphi(v)$.

- (a) Show that a linear transformation $T: V \rightarrow V$ induces a linear transformation $T^*: V^* \rightarrow V^*$, called the **adjoint** of T , such that $\langle T^*\varphi, v \rangle = \langle \varphi, Tv \rangle$ for all $v \in V$ and $\varphi \in V^*$.
- (b) Suppose V is finite-dimensional and choose a basis $v_1, \dots, v_n \in V$. Show that V^* has a unique basis $\varphi_1, \dots, \varphi_n$ with the property that

$$\varphi_i(v_j) = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

- (c) Let A be the matrix representing T with respect to $v_1, \dots, v_n$. Show that the matrix representing T^* with respect to $\varphi_1, \dots, \varphi_n$ is the transpose of the matrix A .
- (d) Show that $T^{**} = T$ under the isomorphism in Problem 6.

Solution. (a) In general, if $T: V \rightarrow W$ is a morphism of F -vector spaces, there is an induced transformation of dual spaces $T^*: W^* \rightarrow V^*$ which sends $\varphi: W \rightarrow F$ to $\varphi \circ T: V \rightarrow F$. By construction, $(T^*\varphi)(v) = (\varphi \circ T)(v) = \varphi(Tv)$ for all $\varphi \in V^*$ and $v \in V$.

(b) The φ_i are uniquely determined by their values on the v_j , so this is only one possible basis $\varphi_1, \dots, \varphi_n$.

(c)

(d)

□

Problem 8. Let $q(x) \in \mathbb{R}[x]$ be an irreducible quadratic polynomial and $e \geq 1$. Find a basis for the $\mathbb{R}$ -vector space $\mathbb{R}[x]/(q(x)^e)$ such that the linear transformation corresponding to x is given by the following matrix:

$$\begin{pmatrix} a & b & & & & \\ -b & a & 1/b & & & \\ & a & b & & & \\ & -b & a & 1/b & & \\ & & & \ddots & & \\ & & & & a & b \\ & & & & -b & a & 1/b \end{pmatrix}$$

Problem 9. Let V be a finite-dimensional F -vector space. Suppose that the minimal polynomial of a linear transformation $T: V \rightarrow V$ can be factored as $(x - \lambda_1)^{e_1} \cdots (x - \lambda_n)^{e_n}$ for distinct elements $\lambda_1, \dots, \lambda_n \in F$. We proved in class that, as $F[x]$ -modules,

$$V \cong V_1 \oplus \cdots \oplus V_n,$$

where $V_i \subseteq V$ is the submodule annihilated by $(x - \lambda_i)^{e_i}$. Show that the resulting coordinate projections $\pi_i: V \rightarrow V_i$ can be written in the form $\pi_i(v) = f_i(T)v$ for polynomials $f_i(x) \in F[x]$.